

Jaehyun Jeon

📍 Seoul, Korea ✉ jaehyunjeon@lgresearch.ai | jaehyunjeon.0126@gmail.com 📞 +82-10-7211-9910

🌐 [Homepage](#) in [LinkedIn](#) 🐙 [Github](#)

Experience

LG AI Research

AI Scientist (EXAONE Lab)

Sep. 2025 – Current

- Contributing to the development of the EXAONE foundation model, focusing on improving Korean language capabilities and knowledge through benchmarks such as the Korean CSAT

NC AI

Research Engineer (DialogTech Team)

Jan. 2026 – Aug. 2026

- Developed and evaluated domain-specialized AI agents for game customer support, collaborating with support teams to define output schemas, field priorities, and an evaluation dataset
- Built an internal eval and trace visualization platform, designed checklist-based eval metrics, and a configurable experiment system, iterating over tool selection and context compression strategies

LG AI Research

Research Engineer Internship (Language Lab)

Sep. 2025 – Dec. 2025

- Built ChatEXAONE Extended Thinking, an agentic feature with integrated web search and patent-specialized workflows tailored to legal document structure, trained on QA-derived trajectory data
- Served as internal design manager, ensuring production-ready service behavior working between model development and UX, including streaming integration and citation rendering

MIR Lab, Yonsei University (Advisor: Youngjae Yu)

Graduate Researcher

Mar. 2024 – Aug. 2025

- Conducted multimodal AI research/projects, focusing on real-world problem solving via model evaluation, alignment, and RAG

Research Internship

Mar. 2023 – Feb. 2024

- Assisted in multimodal AI projects, gaining practical skills in model training/inference at code level and performance analysis

Data Science Lab (DSL), Yonsei University (Student academic society)

Vice President

Jan. 2022 – Dec. 2022

- Society operations: recruitment, networking, session management, project acquisition & progress tracking
- Managed three industry colab projects: contracts and overall project lead (Freewheelin, Pozalabs, VisualCamp)
- Led sessions/study groups on ML topics (Generative models, NLP, MLOps, etc), and contributed to industry/internal projects

Education

Yonsei University

M.S. in Computer Science (GPA: 4.24/4.3, Advisor: Youngjae Yu)

Mar. 2024 – Feb. 2026

B.S. in Computer Science (GPA: 4.26/4.3, Summa Cum Laude, Top 1/106)

Mar. 2018 – Feb. 2024

Publication (◦ Subject / ◦ Contribution)

🎓 [Google Scholar](#) [↗](#)

[1] A1lyn: Aligning LLMs for Accessible Web UI Code Generation (COLM 2026) [pdf](#) [↗](#)

Janghan Yoon, Jaegwan Cho, Junhyeok Kim, Jiwan Chung, **Jaehyun Jeon**, Seungwon Lim, Youngjae Yu

- Developed a GRPO-based framework to align code-generating LLMs with web UI accessibility standards
- Built multi-GPU training pipeline with axe-core-based accessibility compliance reward function design and synthetic datasets

[2] Do MLLMs Capture How Interfaces Guide User Behavior? A Benchmark for Multimodal UI/UX Design Understanding (ACL 2026 Main) [pdf](#) [↗](#) [code](#) [↗](#)

Jaehyun Jeon, Min Soo Kim, Jang Han Yoon, Sumin Shim, Yejin Choi, Hanbin Kim, Dae Hyun Kim[†], Youngjae Yu[†]

- Proposed a multimodal LLM evaluation task for understanding and reasoning about human behavior in UI/UX design
- Built a benchmark using real-world A/B-tested UI pairs with rationales, revealing significant MLLM limitations including near-random selection performance, and outlined future directions for behavior-aware UI generation

[3] Zero-shot Multimodal Document Retrieval via Cross-modal Question Generation (EMNLP 2025 Main) [pdf](#) [↗](#)

Yejin Choi, Jaewoo Park, Janghan Yoon, Saejin Kim, **Jaehyun Jeon**, Youngjae Yu

- Developed a multimodal retrieval framework achieving SOTA performance on out-of-distribution benchmarks

- Proposed preQ generation method and demonstrated improved recall on multilingual Chinese and Korean RAG benchmarks
- [4] **Can Visual Language Models Resolve Textual Ambiguity with Visual Cues? Let Visual Puns Tell You!** (*EMNLP 2024 Main*) [pdf](#) [🔗](#)
- Jiwan Chung, Seungwon Lim, **Jaehyun Jeon**, Seungbeen Lee, Youngjae Yu
- Introduced a benchmark evaluating multimodal literacy in models using images for pun understanding and word disambiguation
 - Developed multimodal pun benchmark through image generation/retrieval and conducted model performance evaluations

Project (◦ Subject / ◦ Contribution)

AI-Powered Patent Specification Drafting Platform via Multi-Agent Pipeline	Apr. 2026 - Sep. 2026
◦ Built a platform for automated patent specification drafting using a multi-agent pipeline with parallel subagents across claim drafting, drawing labeling, and iterative spec generation with escalation logic ◦ Led full-stack and model development, and collaborated iteratively with patent attorneys to define domain-specific rubrics as shared evaluation criteria for human review and model self-assessment loops	
[Yanolja] Automated Accommodation Image Assessment Based on User Preferences (<i>Lead</i>)	Jun. 2025 - Aug. 2025
◦ Developed an end-to-end framework for assessing accommodation images based on human-perceived attractiveness ◦ Built assessment framework identifying informativeness and aesthetics as crucial aspects, created manually labeled datasets, trained and evaluated VLMs, and proposed image sorting strategies to enhance user experience	
[LG Electronics] webOS Platform Development Support Multimodal Chatbot	May. 2024 - Dec. 2024
◦ Built RAG-based multimodal chatbot with webOS knowledge base, automating data preprocessing to answering pipeline ◦ Designed document preprocessing pipeline with markdown conversion, built curated benchmark for performance analysis, implemented reranking to improve in answer quality, and delivered API for usability	
AkaLlama: Korean Specialized 70B LLM Construction link 🔗	May. 2024 - Dec. 2024
◦ Released a 70B LLaMA-based LLM full fine-tuned with curated data to be specialized in Korean ◦ Curated Korean training dataset gathering open-source instruction-following samples through quality analysis and deduplication, and evaluated performance on major Korean benchmarks showing results comparable to leading models	
Visual Distractor: Vision-Language Models Fooled by Images (<i>Lead</i>)	Sep. 2023 - Dec. 2023
◦ Proposed a benchmark measuring VLM hallucination vulnerabilities from visual interference in text question answering ◦ Built datasets by generating diverse questions on anomalous synthetic images and evaluated VLMs for incorrect responses	
User-interactive Image Captioning with Constrained Decoding (<i>Lead</i>) code 🔗	Mar. 2023 - Jun. 2023
◦ Developed an interactive image captioning system enabling users to fix specific words in the generated captions ◦ Implemented constrained decoding for VLMs by modifying Transformers code to enforce fixed words during generation	
[Freewheelin] Establishment of Student Performance Indicators in Service (<i>Lead</i>)	Oct. 2022 - Dec. 2022
◦ Identified students' actual math skill levels to highlight the value of the company's service ◦ Developed an embedding-based model predicting student math performance, enabling personalized study recommendations	

Awards and Honors

Scholarship	
Yonsei-to-Yonsei (Y2Y) Scholarship (Full, merit-based)	Mar. 2024 – Feb. 2026
National Science & Technology Scholarship (Full, merit-based)	Feb. 2022 - Feb. 2024
Academic Excellence Scholarship (Full, merit-based)	Sep. 2018 - Feb. 2022
Awards	
Software Capstone Design Project, Excellence Award (Fall Semester)	Dec. 2023
Software Capstone Design Project, Excellence Award (Spring Semester)	Jun. 2023
KPMG Ideathon Challenge, 3rd Place (Subj: 'Shopping Service with Carbon Emission Prediction')	Feb. 2022
ICPC Sinchon Summer Algorithm Camp Contest, 2nd Place	Aug. 2021

Teaching

Teaching Assistant	
CSI4101, 4102: Software Capstone Design	Sep. 2024 – Dec. 2024
Mentoring	
YDMC (Guiding high school students in career exploration)	Sep. 2023 – Jun. 2024

Skills

Programming

[Language] Python, SQL, C++, JavaScript

[ML/DL & AI] PyTorch, Transformers (Huggingface), vLLM, LangChain

[Tools & Platforms] Linux, AWS, Docker, Git, WandB, MLflow, Figma

Language

Korean (Native), English (Fluent, 1.5 years working with US Army, TOEIC 950, OPIc IH)